

Lab 1: Requirements Engineering & UML Use-Case Modelling

Digital Campus Library Reservation Gateway

Use-Case Flow Specification Report

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Use Case Name	Notify Member of Available Hold
Related Requirement	FR-002 (extends Process Book Return)
Primary Actor	Student Member (recipient); triggered via Head Librarian's Process Book Return
Preconditions	1) A returned book has one or more members waiting in its FIFO hold queue. 2) The Head Librarian has just checked in the returned copy.
Postconditions	Success: The next member in the queue is notified and the copy is reserved for them for 48 hours. Failure: The notification is not delivered, or the copy is released without notifying the intended member.

Main Success Scenario

1. Head Librarian checks in the returned book (base use case: Process Book Return).
2. System detects an active hold queue on the title at the extension point.
3. System identifies the member at position 1 in the FIFO queue.
4. System reserves the physical copy for that member and starts a 48-hour hold timer.
5. System sends an email/SMS notification informing the member the book is ready for pickup.
6. System updates the member's dashboard to show "Ready for Pickup".
7. Use case ends, completing the extension of Process Book Return.

Alternate Flow A1: Member Does Not Collect Within 48 Hours

1. At step 4, the 48-hour hold timer expires before the member checks out the book.
2. System releases the reserved copy back into circulation.
3. System removes the expired member's hold record and notifies them that the hold has lapsed.
4. System repeats steps 3-6 for the next member in the queue, if one exists.
5. Use case ends.